

G-PORT: Guarded Prefix-Oriented Runtime Scheduling for Sparse-Flexible Coded Learning

Research Full Paper

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Abstract

Heterogeneous cloud workers make synchronous distributed learning vulnerable to long-tail iteration latency. Sparse flexible codes reduce redundant work, but they still leave a systems question open: which recoverable row set becomes visible first at runtime. We present G-PORT (Guarded Prefix-Oriented Runtime), a worker-service controller whose coded arm keeps the sparse-flexible code fixed while changing the runtime order in which recoverable row sets are exposed. It combines target-specific row scores, worker speed and delay estimates, deadline-aware matching, exact online decodability checks, and a fixed guard that enables the adaptive arm only when predicted prefix savings exceed overhead; otherwise it falls back to static sparse-flexible placement. We evaluate G-PORT with trace-driven simulation and worker-service prototypes, including TCP, network-stressed TCP, and a direct three-node K3s deployment. Across the worker-service evaluation, G-PORT improves barrier latency by up to 24.8% in deployed K3s stress tests, remains positive across all main TCP, network-stressed TCP, and K3s scale points, and maintains bounded recovery under communication, interference, cancellation, and worker-close stress. These results show that row-exposure order is a practical scheduling lever inside fixed coded-computing deployments.

CCS Concepts

• **Networks** → **Cloud computing**; • **Computer systems organization** → *Distributed architectures*; • **Computing methodologies** → *Distributed computing methodologies*.

Keywords

coded computing, distributed learning, straggler mitigation, cloud scheduling

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1 Introduction

Cloud platforms increasingly run distributed learning jobs on heterogeneous and shared clusters. Worker speeds, queueing delays, network variability, and co-located interference create long-tail iteration times. Tail-at-scale analysis showed that a small fraction of slow components can dominate service latency [1]. Cluster systems turned the same observation into runtime mechanisms: speculative execution mitigates slow MapReduce tasks [2], Mantri diagnoses and reacts to outliers in MapReduce clusters [3], and Dolly uses cloning to reduce straggler impact [4]. These systems established a useful design principle: the scheduler should optimize the unit

that actually gates completion, not just the average work issued to the cluster. For synchronous distributed learning, that gate is revisited at every iteration, so even modest worker-speed mismatch can repeatedly become a barrier-latency problem.

Modern ML cluster schedulers apply this principle at the job and training-stack layers. Optimus models the marginal value of resources for distributed deep learning jobs [5]. Gandiva exploits introspection to time-slice and pack deep learning workloads [6]. Tiresias schedules GPU jobs with learned job progress information [7]. Themis studies fair and efficient GPU sharing [8]. Gavel makes heterogeneity an explicit cluster-scheduling dimension [9]. Pollux co-adapts training batch size and cluster allocation for goodput [10]. Sia extends goodput optimization to heterogeneous ML clusters [11]. Recent systems continue the same trajectory: RL Tune combines learned prioritization with solver-based allocation [12]; Cuckoo optimizes deadline-aware GPU job packing [13]; Sailor automates distributed training over dynamic heterogeneous clusters [14]; What-if analysis diagnoses training stragglers [15]; Holmes localizes irregularities in large GPU fleets [16]; Minder detects faulty machines during large-scale training [17]; GREYHOUND targets fail-slow behavior in hybrid-parallel training [18]; and Zeppelin balances variable-length LLM training work [19]. The common abstraction in these systems is that heterogeneity is managed through job priority, resource allocation, packing, placement, or diagnosis. That abstraction is powerful, but it does not expose the algebraic dependency among encoded rows inside a coded iteration.

Coded computing changes the completion condition of an iteration. Gradient coding uses algebraic redundancy so that a master can recover a full gradient without waiting for every worker [20]. Short-Dot reduces the cost of coded linear transforms through sparse coded products [21]. Coded distributed learning generalizes this idea to machine-learning iterations [22]. Sparse coded matrix multiplication further reduces coded-linear-algebra overhead [23]. Approximate gradient coding studies heterogeneous stragglers under relaxed recovery [24]. Sparse flexible coded learning (SFCL) creates the closest exact baseline for this paper: a fixed sparse-flexible code matrix with multiple legal recovery paths [25]. These constructions move the barrier from “all assigned work is complete” to “some algebraically sufficient row set is complete.” This shift creates a runtime question that code construction alone does not answer: among many legal recovery paths, which one will become visible first under the current worker speeds, dispatch delays, cancellation behavior, and sparse row costs.

The resulting scheduling problem is neither ordinary load balancing nor general cluster placement. In sparse flexible coded learning, the code defines a row-span family rather than a single required worker set [25]. An encoded row is useful only when it arrives with other completed rows whose span contains the target gradient. A cost-balanced runtime can therefore place expensive sparse

rows on fast workers and still delay the first decodable row set, because row cost and row decodability measure different things. A generic heterogeneity-aware runtime can also miss the same signal: cluster schedulers such as Gavel expose heterogeneity as a job-to-resource placement problem [9], whereas coded learning needs a row-prefix objective inside one iteration. This observation gives the background for our method. The scheduler should preserve the fixed sparse-flexible code, predict how a candidate assignment changes the first-decodable prefix, and enable adaptation only when the predicted prefix reduction is large enough to pay for scheduling, extra-work, communication, and cancellation overhead.

We present G-PORT, a guarded worker-service controller for sparse flexible coded learning. The name abbreviates Guarded Prefix-Oriented Runtime: the controller keeps the code fixed and changes which row prefix is exposed first. Its coded arm assigns a target-specific importance score to encoded rows, aggregates scores over row pairs, maps decode-critical pairs to workers predicted to finish early, and runs the same row-span test as the decoder to predict the first-decodable prefix. A fixed online guard enables the adaptive arm only when worker-speed mismatch, predicted first-decode gain, prefix growth, and rows-after-decode counters pass; otherwise G-PORT falls back to static sparse-flexible placement. The design keeps the sparse-flexible code fixed and focuses on the worker-service runtime that controls row exposure order. Coded-only and alternative-fallback variants are treated as ablations, not separate proposed systems.

The paper makes four contributions.

- We identify first-decodable row exposure as a systems objective for sparse flexible coded learning. The objective asks the runtime to expose an algebraically sufficient row prefix early, while leaving the code construction and decoder unchanged.
- We design G-PORT, a guarded worker-service controller with decode-aware row scores, finish-time estimates, first-prefix prediction, and static sparse-flexible fallback. A fixed online guard enables the adaptive coded arm only when the predicted prefix reduction is worth its runtime cost.
- We specify the deployment semantics that make the design safe to use: reassignment changes row-to-worker order but not the recoverable row-span family, decoding is accepted only by the exact row-span test, and guard decisions are made before each iteration rather than after observing the outcome.
- We evaluate the design against scheduler-style baselines using trace-driven simulation and worker-service prototypes, including a direct three-node K3s deployment. Across the main TCP, network-stressed TCP, and K3s worker-service comparisons, G-PORT remains positive at all tested worker counts; its bounded reissue path restores decode success under a single worker-close stress.

2 Background and Motivation

2.1 Coded Distributed Learning

In synchronous distributed learning, the training data are split into m shards and the master coordinates one gradient computation at iteration t . Let w_t be the current model and let $g_j(w_t)$ denote the

partial gradient computed from shard j . The full gradient is

$$g(w_t) = \sum_{j=1}^m g_j(w_t), \quad (1)$$

so an uncoded synchronous system must wait until every shard contribution has arrived. Gradient coding [20] and related coded computation methods [22] avoid this all-worker wait by asking workers to compute encoded linear combinations of shard gradients. The master then reconstructs the same vector $g(w_t)$ after receiving a sufficient subset of encoded results.

2.2 Sparse Flexible Coded Computing

Sparse flexible coded learning constructs a code matrix by stacking two sparse row blocks

$$B = \begin{bmatrix} A^{(1)} \\ A^{(2)} \end{bmatrix} \in \mathbb{R}^{2n \times m}, \quad (2)$$

where n is the number of workers and m is the number of data shards. The matrix in Equation (2) has one column per shard and $2n$ encoded rows. Let $b_r^\top$ be row r of B ; the coefficient b_{rj} says how shard j participates in encoded row r . A zero coefficient means that row r does not touch shard j , which is where sparsity reduces arithmetic work. Let $G_t = [g_1(w_t), \dots, g_m(w_t)]^\top$ denote the stack of shard gradients at iteration t . Encoded row r computes the vector

$$\tilde{g}_r = b_r^\top G_t = \sum_{j=1}^m b_{rj} g_j(w_t). \quad (3)$$

Equation (3) uses $b_r^\top G_t$ as shorthand for the weighted vector sum on the right-hand side. Let $S \subseteq \{1, \dots, 2n\}$ be the indexes of encoded rows that have completed and returned to the master. Define $B_S \in \mathbb{R}^{|S| \times m}$ as the submatrix of B formed by selecting the completed rows in S . The master can decode once

$$\mathbf{1}_m \in \text{span}(B_S^\top). \quad (4)$$

Here $\mathbf{1}_m$ is the all-ones vector of length m , because recovering the full gradient in Equation (1) requires coefficient one on every shard gradient. The notation $\text{span}(B_S^\top)$ denotes the column span of $B_S^\top$, equivalently all linear combinations of the completed row coefficient vectors. Thus Equation (4) says that the completed rows can be linearly combined to produce the all-ones coefficient vector. Equivalently, there exists a decoding vector $c_S \in \mathbb{R}^{|S|}$ such that

$$B_S^\top c_S = \mathbf{1}_m. \quad (5)$$

Multiplying the received encoded gradients by the same coefficients c_S recovers the uncoded full gradient $g(w_t)$.

Decodability is decoder-dependent. Equations (4) and (5) describe exact linear decoding, as implemented by a rank or Gaussian-elimination-style solve. A peeling or BP decoder would use a different success predicate based on whether iterative singleton recovery can finish. Such a predicate can be stricter than the span condition: a returned row set may be linearly sufficient for Gaussian decoding but not peelable by BP. This paper uses the exact row-span predicate throughout; BP-aware first-decodable scheduling is a natural extension with a different decoder predicate.

The sparse flexible construction mainly decides which rows exist and how sparse they are. In a cloud system, the runtime must also

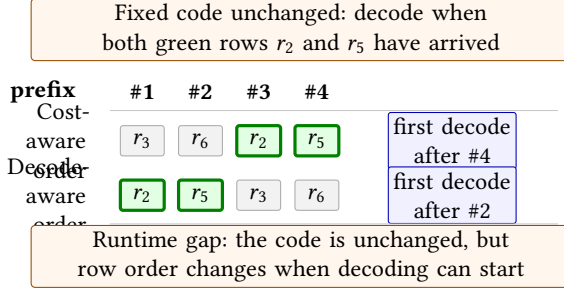


Figure 1: Row completion order changes the first-decodable prefix even when the code is fixed.

decide which row pair goes to which worker. This paper focuses on that runtime degree of freedom.

2.3 Cost-Only Assignment Failure

Sparse encoded rows have different arithmetic costs because they touch different numbers of nonzero data blocks. A cost-aware scheduler assigns expensive rows to fast workers. This can reduce total completed computation time, but the first decodable set may depend on rows that are not locally most expensive. If such rows are delayed, the master still waits even though many other rows have finished.

Figure 1 illustrates this gap with one iteration of a fixed code. The code already makes the pair $\{r_2, r_5\}$ a legal recovery set. A cost-aware order sorts work by local row cost and worker speed, without asking which rows jointly make the target decodable; in the example, it exposes r_3 and r_6 before both recovery rows have arrived. A decode-aware order instead tries to expose a legal recovery set early; here, it places r_2 and r_5 in the first two completions. The runtime gap is therefore the difference between the first decodable times, not a difference in code construction. The guard enables adaptive placement only when the predicted prefix reduction is large enough to dominate scheduling, extra-work, communication, and cancellation costs.

We verify the effect in Figure 1 with a deliberately small mechanism diagnostic before introducing the design. The diagnostic fixes an 8-worker/8-shard sparse-flexible code, generates five code instances, samples 12 worker-state rounds per instance, and changes only the row-pair priority used to expose rows. For each round, it predicts row completion times from sparse row costs and sampled worker speeds, then runs the same row-span check after each completion to find the first-decodable prefix. Static placement has 31.6 ms predicted mean first-decode time. A cost-only priority improves the mean to 25.4 ms, but still needs 9.1 completed rows on average before decode. An oracle decode-priority order, computed by enumerating minimal decodable subsets in this tiny code, reaches 19.3 ms and reduces the first-decodable prefix by 1.2 rows relative to static placement. The oracle is not a deployable scheduler, and the diagnostic does not include network or service overhead; it isolates the reason Figure 1 matters. The design below therefore uses cheap decodability-aware row scores and an online guard rather than treating row cost alone as the scheduling objective.

3 Design

Figure 2 summarizes the system boundary. The sparse flexible code matrix is fixed before an iteration, so the runtime does not change which encoded rows or recovery sets exist. G-PORT operates on the worker-service path: it scores row pairs, predicts whether an adaptive assignment will expose a decodable prefix earlier than static placement, runs a fixed online guard, and then streams rows to TCP or K3s worker services. The exact row-span decoder remains the correctness gate, while the observed first-decode event feeds the guard counters for later rounds.

3.1 First-Decodable-Time Objective

At iteration t , worker j has a random completion time determined by its current speed, queueing delay, and the sparse cost of the assigned row pair. Let $\pi(i)$ denote the worker assigned to row pair i , containing rows i and $i + n$ of the fixed code matrix B . For a time u , let $S_\pi(u)$ be the set of encoded rows completed by time u under assignment π .

Under the exact row-span decoder used in this paper, the first decodable time is

$$\tau(\pi, B) = \inf\{u : \mathbf{1}_m \in \text{span}(B_{S_\pi(u)}^\top)\}. \quad (6)$$

The system objective is

$$\min_{\pi} \mathbb{E}[\tau(\pi, B)]. \quad (7)$$

The objective in Equation (7) differs from minimizing total computation, assigned cost, or mean worker completion time. It cares only about the earliest algebraically sufficient set of returned rows. The exact objective is combinatorial because each completed subset has a different decodability status. We therefore build low-cost surrogates that preserve the central signal: how much a row contributes to recovering the target vector $\mathbf{1}_m$.

3.2 Target-Specific Decode Importance

The scheduler needs a cheap signal for which encoded rows are useful for recovering the target vector $\mathbf{1}_m$. Let $z \in \mathbb{R}^{2n}$ be a row-weight vector over all encoded rows of the fixed code matrix B . If every encoded gradient were available, coefficient z_r would multiply the vector returned by row r . The weighted sum recovers the full gradient exactly when $B^\top z = \mathbf{1}_m$, because then the total coefficient on every shard gradient is one. Among all row-weight vectors that satisfy this recovery condition, we choose the minimum-norm reference decoder

$$z^* = \arg \min_z \|z\|_2^2 \quad \text{s.t.} \quad B^\top z = \mathbf{1}_m. \quad (8)$$

When the target vector is recoverable from the full code matrix, this reference decoder has the closed form

$$z^* = B(B^\top B)^\dagger \mathbf{1}_m. \quad (9)$$

Equations (8) and (9) are computed only from the fixed code matrix and the target full-gradient vector. Here $(\cdot)^\dagger$ is the Moore–Penrose pseudoinverse. The vector z^* is not the online decoder used after partial results arrive; online decoding still solves the exact row-span condition on the returned subset. We use z^* only as a pre-runtime signal: if row r has a large coefficient in this reference decoder,

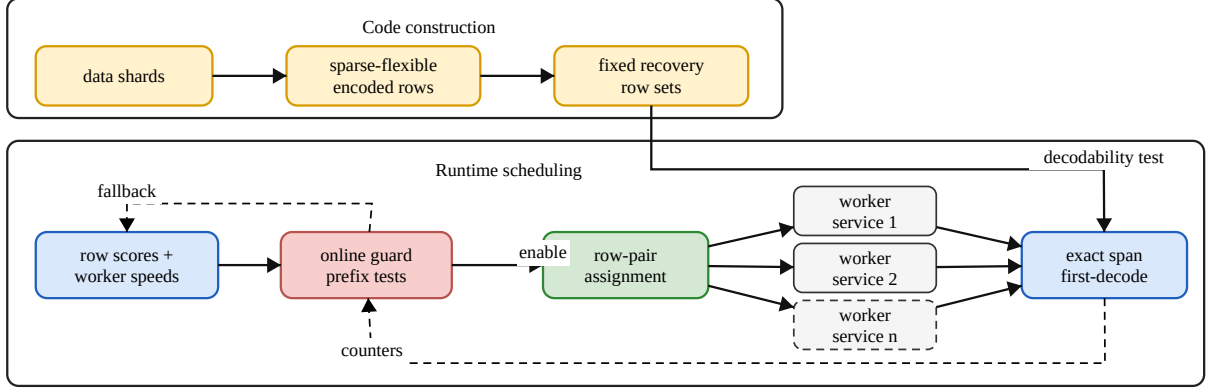


Figure 2: Guarded first-decodable scheduling separates fixed code construction from runtime row exposure.

delaying that row is more likely to delay a recovery path for 1_m . We therefore define the target-specific importance of row r as

$$\rho_r = |z_r^*|. \quad (10)$$

The runtime assigns row pairs rather than individual rows. Pair i contains rows i and $i + n$, and its sparse computation cost is C_i . The scheduler uses the pair priority

$$R_i = (\rho_i + \rho_{i+n})C_i, \quad (11)$$

so a pair becomes high priority only when it is both decode-relevant and costly enough for placement to matter. Equation (11) is a scheduling feature, not a recovery certificate. The runtime still accepts decoding only after the exact row-span test succeeds on streamed completions. Multiplying by C_i is a systems choice rather than a theorem: a high- ρ row with negligible cost does not need a scarce early worker, while a costly row matters only if it is also decode-relevant. The evaluation therefore treats score variants as diagnostics, and the guard remains the protection against regimes where this feature is misaligned with true first-decode criticality.

3.3 Decode-Aware Assignment

The decode-aware scheduler sorts row pairs by decreasing R_i and workers by decreasing estimated speed, then assigns the k th highest-priority row pair to the k th fastest worker. This simple monotone rule requires only the code matrix and a current speed estimate. It is implemented as RANKAWARESPARSEFLEXIBLE in the prototype. The rule is deliberately conservative: it does not change the code, the number of encoded rows, or the algebraic recovery condition. It only permutes the mapping between already-defined row pairs and available workers.

Algorithm 1 shows the full scheduling path. Lines 1–4 compute a target-specific decode-mass signal from the fixed code matrix. Lines 5–7 lift that row-level signal to row pairs and multiply it by sparse row-pair cost, because a high-importance row pair only changes wall-clock time when its computation is nontrivial. Lines 8–12 then perform the monotone placement. The important point is that R_i is not treated as an independent utility score for a completed row. It

Algorithm 1 Decode-aware row-pair assignment

Require: Fixed code matrix B , pair costs $\{C_i\}$, worker speed estimates $\{s_j\}$

Ensure: Row-pair to worker assignment π

```

1: Compute  $z^* = B(B^T B)^\dagger 1_m$ 
2: for all encoded rows  $r$  do
3:    $\rho_r \leftarrow |z_r^*|$ 
4: end for
5: for all row pairs  $i$  do
6:    $R_i \leftarrow (\rho_i + \rho_{i+n})C_i$ 
7: end for
8:  $(i_1, \dots, i_n) \leftarrow$  row pairs sorted by decreasing  $R_i$ 
9:  $(j_1, \dots, j_n) \leftarrow$  workers sorted by decreasing  $s_j$ 
10: for  $k = 1, \dots, n$  do
11:    $\pi(i_k) \leftarrow j_k$ 
12: end for
13: return  $\pi$ 
```

is a proxy for how much delaying that row pair can delay the first completed set whose span contains 1_m . Thus the scheduler tries to make the likely early prefix of completed rows more decodable, rather than merely cheaper or faster in total.

3.4 Deadline-Aware Extension

Decode-aware sorting is the main instantiation of the scheduling idea. As a stress-test extension for runtimes with transient delay, we also define a deadline-aware matching score. Let

$$\widehat{T}_{ij} = d_j + \gamma C_i / s_j \quad (12)$$

be the nominal completion time of pair i on worker j , with delay d_j , speed s_j , and runtime scale γ . For a target deadline q , define the soft completion probability

$$P_{ij}(q) = \exp(-\widehat{T}_{ij}/q). \quad (13)$$

Algorithm 2 Deadline-aware row-pair assignment

Require: Pair scores $\{R_i\}$, costs $\{C_i\}$, speeds $\{s_j\}$, delays $\{d_j\}$, deadline q

Ensure: Row-pair to worker assignment π

```

1: for all row pairs  $i$  and workers  $j$  do
2:    $\hat{T}_{ij} \leftarrow d_j + \gamma C_i / s_j$ 
3:    $P_{ij}(q) \leftarrow \exp(-\hat{T}_{ij}/q)$ 
4:    $W_{ij} \leftarrow R_i P_{ij}(q)$ 
5: end for
6:  $\pi \leftarrow \text{MAXWEIGHTMATCHING}(W)$   $\triangleright$  maximize  $\sum_i W_{i,\pi(i)}$ 
7: return  $\pi$ 

```

The scheduler solves

$$\max_{\pi} \sum_i R_i P_{i,\pi(i)}(q), \quad (14)$$

a standard linear assignment problem. Equations (12), (13), and (14) define the optional deadline-aware policy. It builds $W_{ij} = R_i P_{ij}(q)$ and calls $\text{MAXWEIGHTMATCHING}(W)$, which returns the assignment π maximizing $\sum_i W_{i,\pi(i)}$. In the current prototype, q is the median nominal completion time in the iteration. The extension is useful when transient delay makes the fastest worker by compute speed differ from the earliest worker by response time.

Algorithm 1 requires one target-decoder computation when the code changes and an $O(n \log n)$ sort per scheduling round. The optional deadline-aware variant adds an $n \times n$ weight matrix and a cubic-time exact assignment solve. It is therefore a stress-test policy, not the default path; larger deployments can use the monotone sort, top- k candidate matching, or incremental rematching when few delay estimates change.

3.5 Guard Enablement Conditions

The scheduler is intentionally conditional: it is not expected to improve every iteration. It helps when two conditions hold. First, static placement must be misaligned with decode priority, so high- R_i row pairs are not already on workers likely to complete early. Second, the reassignment must reduce the arrival index of the first decodable set, or at least avoid increasing it by more than the speedup can compensate. In the prototype we expose this through runtime counters for selected rows, completed rows, extra compute, and scheduler time. These counters are used in the evaluation to explain the boundary between strong tail improvements and modest or negative gains.

This yields a fixed counter guard: enable adaptive assignment only when estimated mismatch is positive and recent first-decode counters do not show a larger completed-row prefix. A payload $P = (B, \pi)$ combines the fixed code matrix with a row-to-worker assignment. G-PORT, the main system, compares a safe fallback payload P_0 against candidate payloads and runs an adaptive candidate only if every guard test passes. Its first-decode-aware coded candidate is built by applying the row-pair scores from Algorithm 1 inside the deadline-aware matching of Algorithm 2; we call this coded candidate GUARD-D. If the guard does not pass, G-PORT falls back to static sparse-flexible placement. Rank-only ordering (GUARD-R), coded-only GUARD-D, and alternative fallback choices are retained

as ablations to isolate which part of the system matters; they are not separate proposed methods.

Algorithm 3 is the online guard used by the implemented runtimes and replayed unchanged in diagnostic experiments. In simulation and replay studies, the rule is applied to recorded or synthesized worker states; in multi-process and worker-service runs (TCP, Docker, and K3s), the master executes it before each iteration. The rule has four tests. It first predicts the static fallback payload P_0 and the candidate coded payload P_c , then enables P_c only when worker speeds are heterogeneous enough, the predicted first-decode time is lower, the predicted first-decode prefix does not grow, and recent monitored counters do not show extra prefix growth or post-decode work. Here $\text{cv}(\text{speed})$ is the coefficient of variation of the current worker-speed estimates. Thus the decision is made before the iteration and cannot select a policy after seeing that iteration's latency.

The counter update is conservative. In cold start, the exponential moving-average counters are zero, so the guard is controlled only by the current worker state and the predicted static-vs-candidate prefix comparison. After an enabled round, the monitor updates the counters from the observed first-decode event. After a fallback round, the candidate did not run, so the counters decay toward zero rather than being updated from the fallback's outcome.

PREDICT is an online model over current runtime features, not a replay oracle. For a payload $P = (B, \pi)$, it returns $(\hat{\tau}, \hat{K})$, where $\hat{\tau}$ is predicted first-decode time and $\hat{K}$ is the predicted first-decodable prefix length. Row r assigned to worker $j = \pi(r)$ has nominal service time

$$\hat{x}_{rj} = \beta_d d_j + \beta_c c_r / \max(s_j, \epsilon), \quad (15)$$

where d_j and s_j are the current delay and speed estimates, c_r is row cost, and β_d, β_c are runtime scales. If a worker receives multiple rows, predicted completion is cumulative over its local queue:

$$\hat{t}_r = \sum_{r' \preceq_j r} \max(0, \hat{x}_{r'j}). \quad (16)$$

Here $r' \preceq_j r$ means row r' is no later than r in worker j 's queue. Equations (15) and (16) therefore convert a payload into a predicted row-arrival order. The predictor sorts rows by $\hat{t}_r$ and runs the same incremental row-span test as the decoder; the first successful arrival time is $\hat{\tau}$, and the successful prefix size is $\hat{K}$. Prediction errors are handled conservatively by requiring positive predicted gain, non-increasing predicted prefix, and non-growing EMA prefix counters before enabling the candidate scheduler.

4 Surrogate Analysis

4.1 Recovery Preservation

Lemma 1. For a fixed sparse flexible code B , any row assignment or row permutation preserves the algebraic recovery capability of the code.

Proof sketch. Assignment changes when rows arrive, not the row space generated by all rows. For any permutation matrix P , $\text{rank}(B) = \text{rank}(PB)$ and $\mathbf{1}_m \in \text{span}(B^\top)$ if and only if $\mathbf{1}_m \in \text{span}((PB)^\top)$. Therefore scheduling can improve the first decodable time without weakening the underlying recovery threshold.

Algorithm 3 Online counter guard

Require: Fallback payload P_0 , candidate payload P_c , worker state, thresholds $\theta_{cv}, \theta_g, \theta_K, \theta_a$, EMA counters $\Delta\bar{K}$ and $\bar{A}$, update weight α_{ema} , fallback decay λ

Ensure: Payload for this iteration and updated guard state

- 1: Predict $(\hat{\tau}_0, \hat{K}_0)$ for P_0
- 2: Predict $(\hat{\tau}_c, \hat{K}_c)$ for P_c
- 3: $g \leftarrow (\hat{\tau}_0 - \hat{\tau}_c) / \hat{\tau}_0$
- 4: $\Delta\hat{K} \leftarrow \hat{K}_c - \hat{K}_0$
- 5: $h \leftarrow (\Delta\bar{K} \leq \theta_K) \wedge (\bar{A} \leq \theta_a)$
- 6: **if** $cv(speed) \geq \theta_{cv}$ and $g \geq \theta_g$ and $\Delta\hat{K} \leq \theta_K$ and h **then**
- 7: run candidate scheduler P_c
- 8: observe first-decodable prefix K_c and rows-after-decode A
- 9: $\Delta\bar{K} \leftarrow (1 - \alpha_{ema})\Delta\bar{K} + \alpha_{ema}(K_c - \hat{K}_0)$
- 10: $\bar{A} \leftarrow (1 - \alpha_{ema})\bar{A} + \alpha_{ema}A$
- 11: **else**
- 12: run fallback P_0
- 13: $\Delta\bar{K} \leftarrow \lambda\Delta\bar{K}$; $\bar{A} \leftarrow \lambda\bar{A}$
- 14: **end if**

4.2 Decode-Mass Surrogate

Fix a deadline u . Suppose worker j completes its assigned row pair before u with probability $p_j(u)$, and $p_j(u)$ is monotone in worker speed. A natural surrogate for early decodability is expected completed decode mass:

$$\max_{\pi} \sum_i p_{\pi(i)}(u) R_i. \quad (17)$$

Lemma 2. The optimal assignment for the surrogate in Equation (17) pairs the largest R_i with the largest $p_j(u)$, the second largest with the second largest, and so on.

Proof sketch. The result follows from the rearrangement inequality. For any inversion where $R_a \geq R_b$ but $p_{\pi(a)} < p_{\pi(b)}$, swapping the two assignments changes the objective by $(R_a - R_b)(p_{\pi(b)} - p_{\pi(a)}) \geq 0$. Repeatedly removing inversions gives the monotone assignment.

4.3 Mismatch Creates Scheduling Opportunity

The preceding surrogate also gives a way to state when decode-aware scheduling has room to help. Let $q_j(u) = p_j(u)$ be the probability that worker j finishes before deadline u , and let π_0 be the static row-pair placement. Define the normalized decode-speed mismatch

$$M(\pi_0, u) = \frac{\max_{\pi} \sum_i R_i q_{\pi(i)}(u) - \sum_i R_i q_{\pi_0(i)}(u)}{\max_{\pi} \sum_i R_i q_{\pi(i)}(u) - \min_{\pi} \sum_i R_i q_{\pi(i)}(u)}. \quad (18)$$

In Equation (18), the numerator is the decode-mass headroom left by the static placement π_0 : it compares the best assignment at deadline u with the assignment the fixed runtime would have used. The denominator is the full feasible range between the best and worst assignments, so the ratio normalizes the score to $[0, 1]$ when worker probabilities and row priorities are not both constant. $M = 0$ means the static placement is already optimal for this surrogate; $M = 1$ means it is maximally anti-aligned. The score is an opportunity signal, not a latency guarantee.

Lemma 3. If $M(\pi_0, u) = 0$, no reassignment can improve the expected decode-mass surrogate at deadline u . If $M(\pi_0, u) > 0$, the best monotone reassignment improves the surrogate by exactly $M(\pi_0, u)$ times the surrogate's assignment range.

Proof sketch. The definition subtracts the static surrogate value from the optimal surrogate value and normalizes by the feasible range. Lemma 2 identifies the optimizer as the monotone assignment. Therefore zero mismatch means there is no decode-mass opportunity, while positive mismatch quantifies the available surrogate improvement.

4.4 Prefix Reduction Realizes Latency Gain

The mismatch score says that reassignment has room to help, but it does not by itself imply lower latency. The benefit appears only when a decodable row set arrives earlier. For an assignment π , order returned rows by arrival time and let $T_{\pi, (k)}$ be the time of the k th arrival. Define

$$K_{\pi} = \min\{k : \text{the first } k \text{ arrivals under } \pi \text{ are decodable}\}. \quad (19)$$

Equation (19) defines the first-decodable prefix length: the number of completed rows needed before the row-span decoder can recover the full gradient. If h_{π} is scheduling and monitoring overhead, the first-decode time is

$$\tau_{\pi} = T_{\pi, (K_{\pi})} + h_{\pi}. \quad (20)$$

This gives the guard's decision criterion. Compare static placement π_0 with adaptive placement π . If the candidate does not increase the required prefix ($K_{\pi} \leq K_{\pi_0}$), the predicted gain can be written as

$$\tau_{\pi_0} - \tau_{\pi} = \underbrace{T_{\pi_0, (K_{\pi_0})} - T_{\pi_0, (K_{\pi})}}_{\text{fewer rows waited for}} + \underbrace{T_{\pi_0, (K_{\pi})} - T_{\pi, (K_{\pi})}}_{\text{same prefix arrives earlier}} - h_{\pi}. \quad (21)$$

Equation (21) separates three effects: prefix reduction, faster placement for the same prefix length, and overhead. This is why the guard rejects candidates that predict a longer prefix or too much post-decode work: speed alone is insufficient unless the candidate exposes a decodable prefix early enough to pay for scheduling, communication, and cancellation overhead.

4.5 Surrogate Rationale and Limits

The exact event of interest is whether the rows completed by deadline u form a decodable set. If $\mathcal{D}$ is the family of row subsets that can decode the target vector, then

$$\Pr[\tau(\pi, B) \leq u] = \Pr[S_{\pi}(u) \in \mathcal{D}]. \quad (22)$$

Computing Equation (22) exactly requires reasoning over many completed subsets. The decode-mass surrogate uses a scalar proxy instead: let $Z_{\pi, i}(u)$ indicate whether row pair i finishes before u , and define $X_{\pi}(u) = \sum_i R_i Z_{\pi, i}(u)$. A threshold relaxation treats a row set as promising when $X_{\pi}(u) \geq \theta$. Increasing $\mathbb{E}[X_{\pi}(u)]$ therefore improves this relaxed event when variance does not grow enough to offset the mean gain.

This is a runtime proxy, not a correctness argument. A large R_i suggests that row pair i is useful to expose early, but it does not certify that any returned subset is decodable. The runtime uses the surrogate only to choose an assignment; correctness still comes from the exact online row-span decoder. The guard rejects the

surrogate when predicted prefix or overhead counters indicate that the proxy is unreliable.

5 Experimental Setup

5.1 Workloads and Worker Traces

We use a controlled sparse ridge-regression workload for the main experiments. The generator creates a sparse feature matrix, partitions it into shards, and computes one additive gradient contribution per shard. A successful coded decode therefore recovers the same full gradient as the uncoded shard sum, which matches the linear aggregation model used by gradient coding [20] and sparse flexible coded learning [25]. We use this workload to isolate row-span recoverability, sparse row cost, and first-decode timing, not to claim a full training-stack speedup for all sparse models.

Worker-state streams are controlled stress traces generated by the runtime. Each iteration supplies per-worker speeds, straggler delays, and a slow-worker mask. We use four regimes: stable log-normal heterogeneity, bursty stragglers, drifting base speeds, and phase changes in straggler fraction and slowdown. These regimes model late-task effects in cluster systems [2] and heterogeneous ML scheduling effects [9]. The main SoCC-facing experiments use phase-changing streams because the relative importance of speed, delay, and row cost changes over time. The traces are controlled rather than production logs, so every paired comparison reuses the same worker-state stream across strategies within a run.

5.2 Metrics and Accounting

Recent cluster schedulers report completion time, waiting time or slowdown, resource utilization, and scheduling overhead [12]; distributed-training planners additionally report throughput, cost, and search time [14]. Our setting has one synchronous coded-learning iteration rather than a queue of multi-job training requests, so we adapt these metrics to worker-service rounds. The names below are the names used in the evaluation tables and figures.

- (1) **Mean barrier latency (*Mean (ms)* or *Barrier (ms)*)**. This is the end-to-end synchronous iteration time. It includes first decode, stream cancellation, and worker cleanup, which makes it the main metric for deployed worker-service runs. Stress tables use *Barrier (ms)* for the same quantity.
- (2) **Tail latency (*p95 (ms)*)**. Tail latency is a standard systems metric for cloud services [1]. We report p95 because first-decode savings are useful only if they survive tail workers, cancellation, and communication cleanup.
- (3) **Throughput (*Thr. (iter/s)*)**. Training planners and cluster schedulers commonly report throughput or goodput [10, 14]. We use the reciprocal of barrier latency, reported as completed coded-learning iterations per second.
- (4) **Worker-service overhead (*Post rows and Cancel (ms)*)**. Recent schedulers report scheduling runtime, search time, or planner overhead alongside end-to-end performance [12, 14]. In our tables, *Post rows* is the number of row results completed after the first decodable event but before the round finishes, and *Cancel (ms)* is the cancellation/cleanup time included in the barrier path. Lower values mean less wasted work or less cleanup delay.

- (5) **Correctness and recovery counters (*Succ., Gain, Err., and Recov.*)**. Coded learning requires recovering the intended gradient or row span despite stragglers [20, 25], while distributed training systems also track worker faults and failure reactions [17]. *Succ.* is the decode-success rate. *Gain* is the percentage improvement over the comparison reference stated by the table or figure, such as SFCL in the stress table. *Err.* is closed-worker errors per round, and *Recov.* is bounded row reissues per round.
- (6) **Mechanism and guard diagnostics**. The row-exposure diagnostic reports *Mean gain*, *p95 gain*, and Δ *prefix*, all relative to static placement in the same diagnostic. Guard figures and tables report *decision rate*, *false-enable rate*, *false-disable rate*, *Mean/p95 τ error*, and *Prefix-length error*. Decision rate is the fraction of rounds in which the guard enables adaptive scheduling; false-enable and false-disable rates come from paired replay on the same worker-state stream.

These metrics test whether first-decode prefix savings survive scheduling, communication, and cleanup costs. Operationally, the scheduler should be enabled only when the expected prefix saving exceeds the overhead budget,

$$\Delta T_{\text{prefix}} > T_{\text{sched}} + T_{\text{extra}} + T_{\text{cancel/comm}}.$$

Replication counts, run identifiers, and raw per-run latency traces are artifact details. The main text keeps aggregate barrier, tail, throughput, overhead, decision, and recovery results; the artifact retains the complete run matrix, arm selections, paired comparisons, resource counters, and scripts that regenerate the tables.

5.3 Baselines and External Adapters

The main comparison uses same-runtime baselines so that every method shares the same sparse workload, worker-state stream, serialization path, cancellation logic, and barrier accounting. We use the following comparison points.

- (1) **SFCL**. This is the closest coded-computing baseline: static sparse-flexible coded placement from SFCL [25]. We reimplement it in the same runtime. It fixes the encoded-row assignment before each round and uses neither worker heterogeneity nor first-decode prefix features.
- (2) **RLTune-style adapter**. This adapter follows the systems role of RLTune [12] by selecting among uncoded, speculative, static-coded, and rank-aware coded arms using runtime heterogeneity features. It is a same-runtime adapter, not a port of the full RLTune scheduler.
- (3) **Sailor-style adapter**. This adapter follows the speed/cost modeling role of Sailor [14]. It assigns work using worker-speed and row-cost estimates, but it does not use coded row-span or first-decodable-prefix features.

Guarded enablement. The guard used in the evaluation is a deployment rule over runtime counters, not a selector trained on final latency. The online runtime estimates decode-speed mismatch from row priorities and worker-speed estimates before an iteration starts, predicts if the adaptive policy will shorten the first-decodable prefix, and updates counters after the observed first-decode event. If the predicted prefix is worse, or recent counters show that adaptive

assignment expanded the completed-row prefix, G-PORT rejects the adaptive arm and falls back to static sparse-flexible placement. The portfolio candidates share the same worker-service accounting; the ablations then remove cross-policy execution or change the fallback rule. The `best_safe` ablation chooses the cheaper predicted safe baseline between speed-aware uncoded and static coded placement before the round. Paired safe-baseline replays over logged diagnostics estimate fallback headroom; they do not relabel deployed K3s traces. Unless stated otherwise, the guard thresholds are $\theta_{cv} = 0.20$, $\theta_{g} = 0.01$, $\theta_{K} = 0$ rows, and $\theta_{a} = 1$ row; the EMA update weight is $\alpha_{ema} = 0.30$ and the fallback decay is $\lambda = 0.80$.

5.4 Runtime Prototypes

The experiments use three Python runtimes for sparse coded learning. The labels in Table 1 are configurations of the worker-service runtime: TCP runs local worker services over sockets, TCP+stress adds analytic request/response delay and bandwidth limits to the same TCP path, and K3s runs the same worker endpoint as pods behind headless Services. The simulator and multi-process runtimes support mechanism checks; the main external comparison uses the TCP-family and K3s worker-service realizations.

(1) Event simulator.

The simulator generates sparse ridge data, partitions it into shards, constructs the fixed sparse-flexible code matrix, and samples worker completion times from the configured trace. It runs the exact row-span decoder after each row arrival. We use it for mechanism sweeps, oracle diagnostics, threshold sensitivity, and paired replays because it separates scheduling logic from process, socket, and container overheads.

(2) Multi-process runtime.

The local runtime starts one Python worker process per logical worker. The master assigns encoded row pairs, workers compute sparse encoded gradients over their assigned shard combinations, and completed rows are streamed back to the master. After the first decodable set is reached, the master cancels remaining work and records both first-decode latency and conservative barrier latency, including cancellation cleanup. This runtime validates the master/worker control path without network isolation.

(3) TCP worker-service runtime.

The worker-service runtime launches each worker as an independent TCP service. Master and workers exchange length-prefixed task, row, cancel, stop, and acknowledgement messages. A task message carries the round id, row ids, encoded rows, worker speed/delay state, and current model vector; row messages return encoded gradients, row cost, payload size, and timing counters. The same worker endpoint runs as local services, host-published Docker containers, Docker-bridge containers, and three-node K3s pods reached through headless Services. This validates the worker-service boundary without publishing worker ports. For restricted cloud hosts, the runtime also supports an SSH-forwarded two-node mode. The K3s run is a shared-pod validation, not an isolated production cluster; pod placement, resource knobs, and manifests are artifact metadata. The

runtime also includes the closed-connection stress hook: if a worker disconnects before first decode, the master marks it unavailable, filters unfinished rows that have not arrived, and reissues them to the fastest live worker. Exact decoding remains the correctness test; if reissued rows do not arrive in time, the iteration fails normally and the next round can fall back to SFCL.

The current runtimes are still prototypes: they de-risk the service boundary, cancellation path, and small pod-to-pod TCP deployment, while production-scale cluster scheduling and interference remain future systems work. This is a deployment step rather than a different algorithmic path, because the scheduler only depends on streamed completions and cancellation acknowledgments.

6 Evaluation

We evaluate G-PORT as a runtime scheduling layer, not a new code construction. The evaluation first compares G-PORT with same-runtime external adapters across TCP, stressed TCP, and K3s worker services, then isolates design choices, guard sensitivity, stress recovery, and row-exposure mechanisms. The main comparison reports latency, tail latency, throughput, work overhead, cancellation cost, and decode success; later diagnostics reuse the decision and recovery counters from Section 5.

6.1 Main External Comparison

Table 1 is the main systems result. Every row uses the same payloads, worker-state streams, cancellation path, and barrier accounting inside that runtime. RLTune and Sailor are same-runtime scheduler adapters, not full system ports; they test whether generic heterogeneity-aware scheduling is enough once the runtime and workload are held fixed. The table reports raw metrics rather than normalized gains: mean barrier latency, p95 barrier latency, achieved iteration throughput, post-decode rows, cancellation time, and decode success.

The table makes the runtime boundary explicit: the same adapter set runs under local TCP, stressed TCP, and pod-to-pod K3s services. Generic heterogeneity schedulers improve selected W16/W24 cases but regress at W8 because they do not control which coded rows make the first decodable prefix appear. G-PORT is not always fastest (Sailor wins some K3s W16/W24 speed-dominated cases), but it has the best W8 mean/p95 in all three runtimes, the best stressed-TCP W24 mean/p95, and competitive K3s W24 throughput. The integrated overhead rows show that the gain is not only a latency artifact: G-PORT usually leaves fewer post-decode rows than SFCL and preserves decode success while cancellation time remains included in the barrier path. The artifact keeps completed rows, extra work, dispatch time, bytes, per-seed traces, paired gains, and paired CIs.

6.2 Ablation Study

Table 2 removes deployable pieces of G-PORT in all three runtimes. *w/o FD* removes first-decode-aware row ordering and therefore becomes the Sailor-style speed/cost scheduler. *w/o code* removes the code-aware candidate and becomes the RLTune-style heterogeneity selector. *w/o portfolio* disables the cross-policy wrapper and runs the

Table 1: Main external comparison.

Runtime	Workers	Method	Mean (ms)	p95 (ms)	Thr. (iter/s)	Post rows	Cancel (ms)	Succ.
TCP	W8	SFCL	44.0	80.5	22.7	1.8	16.8	1.00
		RLTune	48.2	76.6	20.8	0.1	16.5	1.00
		Sailor	53.9	77.8	18.6	0.0	19.2	1.00
		G-PORT	40.9	71.5	24.4	1.6	14.1	1.00
	W16	SFCL	38.3	48.3	26.1	10.4	16.5	1.00
		RLTune	36.4	59.1	27.4	1.4	16.5	1.00
		Sailor	35.3	59.5	28.3	0.0	16.3	1.00
		G-PORT	36.6	60.0	27.4	4.1	16.9	1.00
	W24	SFCL	47.5	62.3	21.1	21.9	21.4	1.00
		RLTune	48.3	59.9	20.7	2.8	24.2	1.00
		Sailor	44.3	61.1	22.6	0.0	21.8	1.00
		G-PORT	43.7	61.1	22.9	1.7	20.5	1.00
TCP+stress	W8	SFCL	49.4	88.6	20.2	1.4	15.2	1.00
		RLTune	55.6	86.0	18.0	0.3	17.1	1.00
		Sailor	56.3	89.9	17.8	0.0	15.8	1.00
		G-PORT	48.4	73.9	20.7	1.6	15.2	1.00
	W16	SFCL	53.2	71.0	18.8	9.8	23.6	1.00
		RLTune	53.1	77.0	18.8	1.5	22.5	1.00
		Sailor	53.3	71.2	18.7	0.0	22.2	1.00
		G-PORT	53.1	83.9	18.8	0.0	22.2	1.00
	W24	SFCL	60.9	93.3	16.4	21.0	27.2	1.00
		RLTune	65.4	137.5	15.3	4.1	29.8	1.00
		Sailor	55.5	67.6	18.0	0.0	23.9	1.00
		G-PORT	53.9	61.2	18.5	0.0	23.7	1.00
K3s	W8	SFCL	22.7	60.3	44.0	0.4	1.7	1.00
		RLTune	28.0	55.5	35.7	0.0	1.6	1.00
		Sailor	29.0	54.4	34.5	0.0	1.6	1.00
		G-PORT	19.6	42.0	51.0	0.4	1.7	1.00
	W16	SFCL	14.5	16.3	69.1	3.4	3.6	1.00
		RLTune	9.8	10.9	102.0	1.3	3.2	1.00
		Sailor	9.4	10.2	106.7	0.0	2.9	1.00
		G-PORT	11.3	16.0	88.3	1.5	3.2	1.00
	W24	SFCL	15.7	17.6	63.6	8.9	5.6	1.00
		RLTune	12.8	14.8	78.2	2.5	4.6	1.00
		Sailor	12.4	13.3	80.8	0.0	4.3	1.00
		G-PORT	12.4	13.2	80.5	0.0	4.3	1.00

coded-only GUARD-D arm. Each cell is mean latency / post-decode rows / cancellation time; lower is better.

The ablation shows why the full controller is useful. Speed/cost scheduling is strong when speed dominates (K3s W16/W24) but fragile at W8 and under TCP stress. The heterogeneity selector helps selected counts but has the largest stressed-TCP W24 latency. The w/o-portfolio variant is a clean coded-only ablation and competitive at W8, but it loses the cross-policy fallback that protects G-PORT when uncoded or speed-aware execution is temporarily safer.

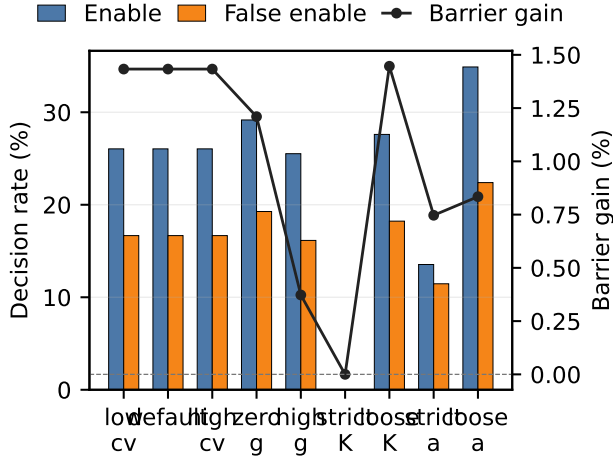
6.3 Guard Sensitivity

Figure 3 replays small threshold changes on paired K3s guard rounds, and Table 3 maps each x-axis label to the guard rule. Each

non-default setting changes one gate while holding the other three at the deployed default. Here theta-cv gates worker-speed heterogeneity, theta-g gates predicted first-decode gain, theta-K gates the candidate-minus-fallback first-decodable prefix delta, and theta-a gates rows of work completed after decode. Lower theta-K or theta-a values are stricter because they allow less prefix growth or post-decode work. Bars report enable decision rate and false-enable rate; the line shows barrier gain. Stricter thresholds reduce false enables but also leave useful opportunities unused, so the default is a conservative middle point rather than a tuned post-hoc optimum.

Table 2: G-P-PORT ablations.

Runtime	Ablation	W8	W16	W24
<i>Cell: mean ms / post rows / cancel ms</i>				
TCP	G-PORT	40.9/1.6/14.1	36.6/4.1/16.9	43.7/1.7/20.5
	w/o FD	53.9/0.0/19.2	35.3/0.0/16.3	44.3/0.0/21.8
	w/o code	48.2/0.1/16.5	36.4/1.4/16.5	48.3/2.8/24.2
	w/o portfolio	43.3/1.6/17.6	38.5/10.6/15.4	47.0/22.1/21.0
TCP+stress	G-PORT	48.4/1.6/15.2	53.1/0.0/22.2	53.9/0.0/23.7
	w/o FD	56.3/0.0/15.8	53.3/0.0/22.2	55.5/0.0/23.9
	w/o code	55.6/0.3/17.1	53.1/1.5/22.5	65.4/4.1/29.8
	w/o portfolio	47.9/2.0/15.3	55.8/11.1/24.8	54.7/19.7/24.6
K3s	G-PORT	19.6/0.4/1.7	11.3/1.5/3.2	12.4/0.0/4.3
	w/o FD	29.0/0.0/1.6	9.4/0.0/2.9	12.4/0.0/4.3
	w/o code	28.0/0.0/1.6	9.8/1.3/3.2	12.8/2.5/4.6
	w/o portfolio	20.3/0.5/1.7	14.2/3.6/3.6	15.9/9.1/5.7

**Figure 3: Guard-threshold sensitivity.****Table 3: Figure 3 settings.**

Label	theta-cv	theta-g	theta-K	theta-a
low theta-cv	0.10	0.01	0	1
default	0.20	0.01	0	1
high theta-cv	0.35	0.01	0	1
zero theta-g	0.20	0.00	0	1
high theta-g	0.20	0.03	0	1
strict theta-K	0.20	0.01	-1	1
loose theta-K	0.20	0.01	1	1
strict theta-a	0.20	0.01	0	0
loose theta-a	0.20	0.01	0	2

Table 4: K3s W24 recovery counters.

Stress	Method	Succ.	Barrier	Gain	Err.	Recov.	Cancel
			(ms)		/round	/round	(ms)
Base	SFCL	1.00	16.6	0.0%	0.00	0.00	5.7
Base	G-PORT	1.00	12.6	23.7%	0.00	0.00	4.4
Base	Guard-D	1.00	16.6	-0.3%	0.00	0.00	5.6
Cancel+20	SFCL	1.00	33.7	0.0%	0.00	0.00	24.8
Cancel+20	G-PORT	1.00	32.0	5.1%	0.00	0.00	23.6
Cancel+20	Guard-D	1.00	33.9	-0.7%	0.00	0.00	24.7
CPU hog	SFCL	1.00	16.6	0.0%	0.00	0.00	5.6
CPU hog	G-PORT	1.00	12.5	24.8%	0.00	0.00	4.3
CPU hog	Guard-D	1.00	16.5	0.6%	0.00	0.00	5.5
Close	SFCL	1.00	16.7	0.0%	1.00	0.00	5.5
Close	G-PORT	0.67	10.8	35.0%	1.00	0.00	2.8
Close	Guard-D	1.00	16.6	0.1%	1.00	0.00	5.5
Close+R	SFCL	1.00	16.8	0.0%	1.00	1.00	5.7
Close+R	G-PORT	1.00	12.7	24.2%	1.00	0.33	4.4
Close+R	Guard-D	1.00	16.8	0.2%	1.00	1.00	5.7

6.4 Stress and Recovery

Table 4 reports K3s W24 stress tests using the correctness and recovery counters from Section 5. *Stress* is the injected runtime condition, not a new workload: Base has no injection, Cancel+20 delays cancellation, CPU hog adds co-located interference, Close closes one worker connection, and Close+R repeats Close with bounded reissue. Succ. is decode success; Gain is relative to SFCL in the same stress; Err. and Recov. are closed-worker errors and bounded reissues per round.

The cases separate cleanup, interference, and connection-loss effects. CPU hog keeps the gain, while delayed cancellation shrinks it to 5.1%. Close without recovery is only a diagnostic failure mode: fast successful rounds but 67% decode success. Bounded reissue

Table 5: Offline row-exposure replay.

Priority	FD mean (ms)	FD p95 (ms)	Mean gain	p95 gain	Δ prefix
Static	31.6	117.0	0.0%	0.0%	0.0
Cost-only	25.4	60.8	19.5%	48.1%	-0.4
Runtime score	32.8	85.6	-3.8%	26.8%	0.0
Oracle-minset	19.3	52.6	38.9%	55.1%	-1.2
Random	38.2	141.2	-21.1%	-20.7%	+0.5

restores success to 1.0 and keeps a 24.2% gain, so we claim single-connection recovery semantics rather than production fault tolerance.

6.5 Mechanism: Row Exposure

Table 5 is an offline 8-worker/8-shard replay, not a TCP or K3s deployment. It fixes the code and worker states, changes only row-pair priority, and measures predicted first-decode time. The oracle is not deployable; it only shows the headroom from exposing decode-important rows earlier. The cheap runtime score helps p95 but not mean latency, so G-PORT treats row score as a guarded runtime feature.

Full row-score sweeps, sampled-minset variants, score-build times, and per-seed ranges are artifact material. Among the five priorities, G-PORT uses the deployable *Runtime score* priority inside its Guard-D coded arm; Static, Cost-only, Oracle-minset, and Random are controls that respectively represent the SFCL baseline, a cost-only scheduler, a nondeployable upper-bound oracle, and a sanity lower bound. This diagnostic supports a narrow claim: decodability features create useful candidates, and the online guard decides when to trust the runtime score.

6.6 Limitations

This is a prototype validation: multi-process runs measure sparse kernels; TCP adds serialization and cancellation; network stress adds analytic RTT/bandwidth; Docker and K3s validate containerized worker-service paths at local and small multi-node scales. The K3s pods do not reserve exclusive CPU or memory. The artifact keeps pod placement, events, resource-counter snapshots when available, and Docker stress hooks for slow cancellation, closed TCP connections, and worker exits. The prototype handles one closed connection by bounded row reissue, but production scheduling, interference control, replicated master state, multi-failure recovery, distributed cancellation, larger sparse workloads, adaptive selectors, and approximate decoders remain future work.

7 Related Work

Coded computation and gradient coding. Coded computation uses algebraic redundancy to ignore delayed workers in linear transforms [22] and gradients [20]. Short-Dot [21] and coded sparse matrix multiplication [23] reduce sparse coded-linear-algebra overhead, and approximate gradient coding further trades exactness for communication and recovery cost [24]. These works decide what redundancy to create, how many workers are needed, or how

expensive decoding is once results arrive. Our problem is different: the sparse-flexible code matrix is fixed before a round starts, and the runtime controls which legal rows become visible to the decoder first. This separates code construction from code-aware row exposure.

Sparse and flexible coded learning. Zhang, He, and Dai [25] propose the closest baseline: a sparse flexible code whose valid recovery combinations make placement matter. Two assignments can have the same recovery capability but different first-decode times because the decoder observes rows in the order induced by worker speeds and row assignments. Static sparse-flexible placement therefore provides the recoverable row-span family, while our scheduler changes only the runtime mapping from row-pairs to workers. The exact decoder remains the correctness test; scheduling affects latency by reducing the prefix that is likely to be decodable, not by changing the algebraic code.

Cluster scheduling for heterogeneous ML. Large-scale ML schedulers attack heterogeneity at the job, GPU, or cluster allocation layer. Optimus [5], Gandiva [6], Tiresias [7], Themis [8], Gavel [9], Pollux [10], and Sia [11] optimize placement, fairness, migration, or resource scaling for distributed training jobs. RL Tune [12] and Cuckoo [13] continue this line with recent scheduling policies for heterogeneous cloud and GPU-cluster workloads. These systems choose where and when jobs or training tasks run. G-PORT works at a lower granularity inside one coded-learning iteration: it assumes the worker pool and code matrix are given, then orders encoded row exposure so that the first recoverable prefix arrives earlier.

Dynamic training runtimes and fail-slow diagnosis. Recent ML-systems work also adapts to runtime variation within a job. Sailor [14] targets dynamic heterogeneous training, while Zeppelin [19] balances variable-length LLM training work. Complementary diagnostic systems and analyses characterize stragglers and fail-slow behavior, including What-if analysis [15], Holmes [16], Minder [17], and GREYHOUND [18]. Their main unit is a job, task, replica, or observed failure mode. Our unit is an encoded row-pair whose usefulness is coupled through the code matrix. This coupling is why a speed-only or cost-only priority can improve local work time but still expose a late decodable prefix.

Classic straggler mitigation. Cloud systems have long used tail-tolerant services [1], speculation [2], outlier management [3], and cloning [4] to mask slow tasks. These mechanisms treat completed task outputs as independently useful and often pay extra work to reduce tail latency. Sparse-flexible coded learning changes the runtime objective: the master can stop only when the completed encoded rows span a valid recovery set. G-PORT therefore complements speculation and cluster-level scheduling by making the redundancy already present in the code more visible to the decoder before adding more work.

8 Conclusion

Fixed sparse-flexible codes define recovery paths; G-PORT decides which path becomes visible first. It exposes decode-critical row pairs earlier when the guard predicts a prefix benefit and otherwise

falls back to static placement. Across the worker-service experiments, mismatch creates an opportunity, first-decodable prefix reduction turns it into tail gain, and the guard avoids many overhead-dominated regimes. K3s validates the TCP worker-service path; production scheduling, full recovery, and end-to-end sparse learning remain open.

Artifact Availability

The anonymous artifact is available at <https://anonymous.4open.science/r/G-port-4FD5/>.

References

- [1] Jeffrey Dean and Luiz André Barroso. 2013. The Tail at Scale. *Commun. ACM* 56, 2 (2013), 74–80. doi:10.1145/2408776.2408794
- [2] Matei Zaharia, Andy Konwinski, Anthony D. Joseph, Randy H. Katz, and Ion Stoica. 2008. Improving MapReduce Performance in Heterogeneous Environments. In *Proceedings of the 8th USENIX Symposium on Operating Systems Design and Implementation*. USENIX Association, San Diego, CA, 29–42.
- [3] Ganesh Ananthanarayanan, Srikanth Kandula, Albert Greenberg, Ion Stoica, Yi Lu, Bikas Saha, and Edward Harris. 2010. Reining in the Outliers in Map-Reduce Clusters using Mantri. In *9th USENIX Symposium on Operating Systems Design and Implementation*. USENIX Association, Vancouver, BC, 265–278.
- [4] Ganesh Ananthanarayanan, Ali Ghodsi, Scott Shenker, and Ion Stoica. 2013. Effective Straggler Mitigation: Attack of the Clones. In *10th USENIX Symposium on Networked Systems Design and Implementation*. USENIX Association, Lombard, IL, 185–198.
- [5] Yanghua Peng, Yixin Bao, Yangrui Chen, Chuan Wu, and Chuanxiong Guo. 2018. Optimus: An Efficient Dynamic Resource Scheduler for Deep Learning Clusters. In *Proceedings of the Thirteenth European Conference on Computer Systems (EuroSys '18)*. ACM, Porto, Portugal, 1–14. doi:10.1145/3190508.3190517
- [6] Wencong Xiao, Romil Bhardwaj, Ramachandran Ramjee, Muthian Sivathanu, Nipun Kwatra, Zhenhua Han, Pratyush Patel, Xuan Peng, Hanyu Zhao, Quanlu Zhang, Fan Yang, and Lidong Zhou. 2018. Gandiva: Introspective Cluster Scheduling for Deep Learning. In *Proceedings of the 13th USENIX Symposium on Operating Systems Design and Implementation (OSDI '18)*. USENIX Association, Carlsbad, CA, 595–610.
- [7] Juncheng Gu, Mosharaf Chowdhury, Kang G. Shin, Yibo Zhu, Myeongjae Jeon, Junjie Qian, Hongqiang Liu, and Chuanxiong Guo. 2019. Tiresias: A GPU Cluster Manager for Distributed Deep Learning. In *Proceedings of the 16th USENIX Symposium on Networked Systems Design and Implementation (NSDI '19)*. USENIX Association, Boston, MA, 485–500.
- [8] Kshitij Mahajan, Arjun Balasubramanian, Arjun Singhvi, Shivaram Venkataraman, Aditya Akella, Amar Phanishayee, and Shuchi Chawla. 2020. Themis: Fair and Efficient GPU Cluster Scheduling. In *Proceedings of the 17th USENIX Symposium on Networked Systems Design and Implementation (NSDI '20)*. USENIX Association, Santa Clara, CA, 289–304.
- [9] Deepak Narayanan, Keshav Santhanam, Fiodar Kazhemiaka, Amar Phanishayee, and Matei Zaharia. 2020. Heterogeneity-Aware Cluster Scheduling Policies for Deep Learning Workloads. In *Proceedings of the 14th USENIX Symposium on Operating Systems Design and Implementation (OSDI '20)*. USENIX Association, Virtual Event, 481–498.
- [10] Aurick Qiao, Sang Keun Choe, Suhas Jayaram Subramanya, Willie Neiswanger, Qirong Ho, Hao Zhang, Gregory R. Ganger, and Eric P. Xing. 2021. Pollux: Co-adaptive Cluster Scheduling for Goodput-Optimized Deep Learning. In *Proceedings of the 15th USENIX Symposium on Operating Systems Design and Implementation (OSDI '21)*. USENIX Association, Virtual Event, 1–18.
- [11] Suhas Jayaram Subramanya, Daiyaan Arfeen, Shouxu Lin, Aurick Qiao, Zhihao Jia, and Gregory R. Ganger. 2023. Sia: Heterogeneity-Aware, Goodput-Optimized ML-Cluster Scheduling. In *Proceedings of the 29th Symposium on Operating Systems Principles (SOSP '23)*. ACM, Koblenz, Germany, 642–657. doi:10.1145/3600006.3613175
- [12] Shruti Dongare, Redwan Ibne Seraj Khan, Hadeel Albahar, Nannan Zhao, Diego Meléndez-Maita, and Ali R. Butt. 2025. Hybrid Learning and Optimization-Based Dynamic Scheduling for DL Workloads on Heterogeneous GPU Clusters. In *Proceedings of the ACM Symposium on Cloud Computing (SoCC '25)*. ACM, Online, USA, 557–570. doi:10.1145/3772052.3772257
- [13] Yuzheng Zhang, Renyu Yang, Junhong Liu, Weihang Jiang, Tianyu Ye, Yiqiao Liao, Penghao Zhang, Tiezi Zhang, Kun Shang, Tianyu Wo, Chunming Hu, Chengru Song, and Jin Ouyang. 2025. Cuckoo: Deadline-Aware Job Packing on Heterogeneous GPUs for DL Model Training. In *Proceedings of the ACM Symposium on Cloud Computing (SoCC '25)*. ACM, Online, USA, 847–859. doi:10.1145/3772052.3772266
- [14] Foteini Strati, Zhendong Zhang, George Manos, Ixeia Sánchez Pérez, Qinghao Hu, Tiancheng Chen, Berk Buzcu, Song Han, Pamela Delgado, and Ana Klimovic. 2025. Sailor: Automating Distributed Training over Dynamic, Heterogeneous, and Geo-distributed Clusters. In *Proceedings of the ACM SIGOPS 31st Symposium on Operating Systems Principles (SOSP '25)*. ACM, Seoul, Republic of Korea, 17 pages. doi:10.1145/3731569.3764839
- [15] Jinkun Lin, Ziheng Jiang, Zuquan Song, Sida Zhao, Menghan Yu, Zhanghan Wang, Chenyuan Wang, Zuocheng Shi, Xiang Shi, Wei Jia, Zherui Liu, Shuguang Wang, Haibin Lin, Xin Liu, Aurojit Panda, and Jinyang Li. 2025. Understanding Stragglers in Large Model Training Using What-if Analysis. In *Proceedings of the 19th USENIX Symposium on Operating Systems Design and Implementation (OSDI '25)*. USENIX Association, Boston, MA, 483–498.
- [16] Zhiyi Yao, Pengbo Hu, Congcong Miao, Xuya Jia, Zuning Liang, Yuedong Xu, Chunzhi He, Hao Lu, Mingzhuo Chen, Xiang Li, Zekun He, Yachen Wang, Xian-neng Zou, and Junchen Jiang. 2025. Holmes: Localizing Irregularities in LLM Training with Mega-scale GPU Clusters. In *Proceedings of the 22nd USENIX Symposium on Networked Systems Design and Implementation (NSDI '25)*. USENIX Association, Philadelphia, PA, 523–540.
- [17] Jiangwei Deng, Tong Shi, Haibin Zhou, Xing Yang, Zhigang Zhao, Leizhen Zhang, Peng Xiao, Chuan Ma, Peng Wang, Meilong Hu, Yuxin Liu, Yongjie Ye, Zhi Wang, and Gaopeng Yang. 2025. Minder: Faulty Machine Detection for Large-scale Distributed Model Training. In *Proceedings of the 22nd USENIX Symposium on Networked Systems Design and Implementation (NSDI '25)*. USENIX Association, Philadelphia, PA, 505–521.
- [18] Tianyuan Wu, Wei Wang, Yinghao Yu, Siran Yang, Wenchao Wu, Qinkai Duan, Guodong Yang, Jiamang Wang, Lin Qu, and Liping Zhang. 2025. GREYHOUND: Hunting Fail-Slows in Hybrid-Parallel Training at Scale. In *Proceedings of the USENIX Annual Technical Conference (USENIX ATC '25)*. USENIX Association, Boston, MA, 731–747.
- [19] Chang Chen, Tiancheng Chen, Jiangfei Duan, Qianchao Zhu, Zerui Wang, Qinghao Hu, Peng Sun, Xiuhong Li, Chao Yang, and Torsten Hoefler. 2026. Zeppelin: Balancing Variable-Length Workloads in Data Parallel Large Model Training. In *Proceedings of the European Conference on Computer Systems (EuroSys '26)*. ACM, Edinburgh, United Kingdom, 1879–1893. doi:10.1145/3767295.3769369
- [20] Rashish Tandon, Qi Lei, Alexandros G. Dimakis, and Nikos Karampatziakis. 2017. Gradient Coding: Avoiding Stragglers in Distributed Learning. In *Proceedings of the 34th International Conference on Machine Learning (Proceedings of Machine Learning Research, Vol. 70)*. PMLR, Sydney, Australia, 3368–3376.
- [21] Sanghamitra Dutta, Viveck R. Cadambe, and Pulkrit Grover. 2016. Short-Dot: Computing Large Linear Transforms Distributedly Using Coded Short Dot Products. In *Advances in Neural Information Processing Systems 29*. Curran Associates, Inc., Barcelona, Spain, 2092–2100.
- [22] Kangwook Lee, Maximilian Lam, Ramtin Pedarsani, Dimitris Papailiopoulos, and Kannan Ramchandran. 2018. Speeding Up Distributed Machine Learning Using Codes. *IEEE Transactions on Information Theory* 64, 3 (2018), 1514–1529. doi:10.1109/TIT.2017.2736066
- [23] Sinong Wang, Jiashang Liu, and Ness B. Shroff. 2018. Coded Sparse Matrix Multiplication. In *Proceedings of the 35th International Conference on Machine Learning (Proceedings of Machine Learning Research, Vol. 80)*. PMLR, Stockholm, Sweden, 5152–5160.
- [24] Heekang Song and Wan Choi. 2025. Approximate Gradient Coding for Distributed Learning with Heterogeneous Stragglers. In *Advances in Neural Information Processing Systems 38 (NeurIPS '25)*. Curran Associates, Inc., San Diego, CA. <https://openreview.net/forum?id=QIDyoo8qLY>
- [25] Jingzhe Zhang, Xiaofan He, and Huaiyu Dai. 2025. Speeding Up Distributed Learning via Sparse and Flexible Coded Computing. *IEEE Transactions on Information Theory* 71, 4 (2025), 3167–3180. doi:10.1109/TIT.2024.3510293